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# Sunghwan Ko

Ph.D. Student in Mathematics, Seoul National University

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## RESEARCH INTERESTS

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- Coxeter polytopes and reflection groups
- Convex real projective structures and their deformation spaces
- Hilbert geometry and higher Teichmüller theory
- Discrete subgroups of Lie groups

## EDUCATION

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### Seoul National University

Ph.D. in Mathematics

Advisor: Gye-Seon Lee

Mar 2021-Feb 2027

(expected)

### Sogang University

B.Sc. in Mathematics

Mar 2015-Feb 2021

## RESEARCH

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### PREPRINT

- Finiteness of integral representations on 2-perfect truncation polytopes. [arXiv:2604.22243](https://arxiv.org/abs/2604.22243), 2026

### WORKS IN PROGRESS

- Infinitesimal deformations of reflection groups and their orientation-preserving subgroups. With Eun-Gyu Jang
- Integral representations on compact Coxeter polygons. With Seong Youn Kim

## TALKS

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Workshop for Young Mathematicians in Korea

Finiteness of integral representations on 2-perfect truncation polytopes · 19 Aug 2026

3rd KIAS Graduate Student Meeting in Pure Mathematics, KIAS

Infinitesimal deformations of representations and group cohomology · 7 May 2026

The 21st East Asian Conference on Geometric Topology, Ewha Womans University

Finiteness of integral Vinberg representations for Coxeter truncation polytopes · 4 Feb 2026

Dynamical Group Theory VII, Seoul National University (Lightning talk)

Finiteness of integral Vinberg representations for truncation polytopes · 26 Nov 2025

Exchange Seminar, Seoul National University

Finiteness of integral representations arising from Coxeter truncation polytopes · 13 Nov 2025

Séminaire Groupes et Actions, Laboratoire de Mathématiques d'Orsay

Finiteness of integral representations coming from truncation polytopes · 12 May 2025

Exchange Seminar, Seoul National University

Finiteness of integral representations of Coxeter 4-gons · 26 Sep 2023

## CONFERENCES AND WORKSHOPS

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- Workshop for Young Mathematicians in Korea, KAIST, 19-21 Aug 2026
- 3rd KIAS Graduate Student Meeting in Pure Mathematics, KIAS, 7 May 2026
- Homogeneous Dynamics and Anosov Representations, Simons Laufer Mathematical Sciences Institute, virtual, 20-24 Apr 2026
- Teichmüller Theory and Related Topics, Fillmore Grand Binhu, Hefei, 4-10 Apr 2026
- The 21st East Asian Conference on Geometric Topology, Ewha Womans University, 2-5 Feb 2026

CONFERENCES AND WORKSHOPS - CONTINUED

- Introductory Workshop: Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups, Simons Laufer Mathematical Sciences Institute, 26-30 Jan 2026
- Pathways Workshop: Topological and Geometric Structures in Low Dimensions & Geometry and Dynamics for Discrete Subgroups of Higher Rank Lie Groups, Simons Laufer Mathematical Sciences Institute, 21-23 Jan 2026
- Dynamical Group Theory VII, Seoul National University, 26-28 Nov 2025
- One-Day Festival of Group Actions, Geometry and Dynamics, KIAS, Seoul, 6 Aug 2025
- KAIST-KIAS Geometry and Topology Exploration Workshop I, KIAS, 10-13 Jun 2025
- Higher Rank Geometric Structures, Higgs Bundles and Physics, IHP, Paris, 19-27 May 2025
- Geometric Structures and Discrete Group Actions, Higher Rank Geometric Structures program, 14 Apr-11 Jul 2025
- Teichmüller Theory and Beyond, Seoul National University, 17-21 Feb 2025
- Workshop on Higher Teichmüller Theory, Chern Institute of Mathematics, Tianjin, 15-24 May 2024

TEACHING EXPERIENCE

|                                          |             |
|------------------------------------------|-------------|
| Seoul National University                |             |
| Topics in Topology                       | Fall 2026   |
| Introduction to Differential Geometry    | Spring 2026 |
| Calculus Practice II                     | Fall 2025   |
| Introduction to Linear Algebra           | Spring 2025 |
| Introduction to Linear Algebra           | Fall 2024   |
| Calculus Practice I                      | Spring 2024 |
| Calculus Practice II                     | Fall 2023   |
| Calculus Practice I                      | Spring 2023 |
| Introduction to Differential Geometry II | Fall 2022   |
| Calculus Practice I                      | Spring 2022 |
| Calculus Practice II                     | Fall 2021   |
| Calculus Practice I                      | Spring 2021 |

AWARDS AND HONORS

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|-----------------------------------------------------------------------------------------------------|-------------|
| Outstanding TA (Calculus I), Seoul National University                                              | Spring 2021 |
| Dean's List, Sogang University                                                                      | Feb 2021    |
| Bronze Award, Korean Contest of Mathematics for Undergraduate Students, Korean Mathematical Society | Dec 2018    |

MILITARY SERVICE

Completed mandatory military service, Republic of Korea Army · Apr 2017-Dec 2018